DESIGN COMPUTED · FREE GUIDE

Make people feel the forces: the human cantilever and the Forth Bridge

The free guide to the reel on the Forth Bridge: why it was built after the Tay Bridge collapse, how a living model explained tension and compression in 1887, the numbers behind the steel, and a chair-and-ruler experiment to feel a cantilever yourself.



A bridge built under suspicion

The Forth Bridge, near Edinburgh, was opened on 4 March 1890 by the Duke of Rothesay, the future Edward VII. It was designed by Sir John Fowler and Sir Benjamin Baker and was the first major steel structure in Britain.

After the Tay Bridge collapsed in 1879, confidence in the original Forth design by Thomas Bouch dried up and work stopped. The inquiry found the Tay Bridge 'badly designed, badly constructed and badly maintained', and Bouch's Forth design had assumed a wind load far below what the Board of Trade later required.

The living model

In 1887, to illustrate how the cantilevers carry load, the Japanese engineer Kaichi Watanabe sat in a chair supported between Fowler and Baker. The poles show compression, the ropes show tension, and bricks at the far ends act as counterweights.

A cantilever is a structure that sticks out from a support and is balanced by a counterweight on the other side. Forth's three double cantilevers reach out from their towers until they meet short suspended girders between them.

The numbers

Opened	4 March 1890.
Steel	About 55,000 tonnes.
Rivets	6.5 million.
Main spans	About 520 m each (reported between 1,700 and 1,709 ft).
Total length	2,467 m.
Workforce	About 4,600 workers at the peak; 57 deaths recorded in 1890, with later counts of 73 to 78.

Try it yourself: feel a cantilever

1. Hold a 30 cm ruler flat on a table with a book on the end that stays on the table: the long end sticks out over the edge.
2. Push the ruler out until it tips: the book is the counterweight. Find the longest overhang that still holds.
3. Add a second book on the overhanging end and see how much you must slide back the ruler to balance it.
4. Now build a double cantilever: two rulers, two books, a short card bridge between the ends.
5. Press on the card with your finger and feel which parts push (compression) and which pull (tension).

The rule to remember

If people can't picture a structure, let them feel its forces: a demonstration beats a calculation they cannot follow.

Default answer	Show equations to people who fear the structure.
Form-finder answer	Make the forces visible with something they can see or hold.
Check	Ask: what is the human model for the load path in my project?

Go further

Forth Bridge (Wikipedia)
https://en.wikipedia.org/wiki/Forth_Bridge

Cantilever bridge (Wikipedia)
https://en.wikipedia.org/wiki/Cantilever_bridge

UNESCO: The Forth Bridge
https://whc.unesco.org/en/list/1485

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Historic photographs: Philip Phillips, 1880s, public domain.
Facts: Wikipedia (Forth Bridge).

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